

Tanishq Bhushan Chaudhari

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Education

- **B.Tech. in Computer Science and Engineering** 2023 – 2027
Indian Institute of Technology Gandhinagar (IIT Gandhinagar) CPI: 8.51
3rd Year Undergraduate | Relevant Coursework: Data Structures & Algorithms, OOP & Design, OS, Computer Networks, Databases

Projects

- **ChatForge - Real-Time Chat Application** (Link)
 - Built a **Node.js/Express** backend with **MongoDB** models for users, conversations, and messages, enabling protected one-to-one chat APIs with modular **Service-Oriented Architecture** across routes, controllers, and socket handlers.
 - Implemented a **multi-layered security framework**: JWT authentication, bcrypt password hashing, socket-level auth middleware, and secure route protection for all user workflows.
 - Developed **real-time messaging** using **Socket.io** with room-based delivery, typing indicators, read receipts, and persistent storage; designed timestamp-based pagination for large-scale chat history retrieval.
 - Added **Redis in-memory** presence tracking for online/offline state; integrated **Kafka** producer-consumer pipelines for asynchronous, scalable event processing and logging.
 - Containerized using **Docker & docker-compose**; wrote **Jest/Supertest** tests covering auth, messaging, and protected endpoint workflows.
- **QuickBites - Food Delivery Backend System** (Link)
 - Designed a normalized **MySQL** schema with **10+ entities**, integrity constraints, and optimized SQL queries with **EXPLAIN**-based profiling for search, menu, and order workflows.
 - Implemented a **custom B+ Tree index in Python** supporting insert, delete, exact search, and range queries on **1M+** records; benchmarked against brute-force across multiple workloads.
 - Built **REST APIs** with **RBAC**, session-based authentication, audit logging, and **cryptographically secure** access control for admin and user operations.
 - Implemented **ACID transaction management**, rollback, and crash recovery; tested **multithreaded concurrent workloads**, race conditions, and failure scenarios to validate isolation under load.
 - Added **hash-based sharding** with query routing across multiple simulated shards in **Docker** containers, supporting **horizontal scaling** and high-throughput backend extensibility.

Skills

- **Programming:** C++, Java, Python, SQL, JavaScript/Node.js (*proficient in OOP design patterns, applied across academic and project work*)
- **Backend & System Design:** Node.js, Express.js, REST API Design, Service-Oriented Architecture (SOA), JWT Auth, Socket.io/WebSockets, MongoDB
- **Databases & In-Memory Systems:** MySQL, PostgreSQL, MongoDB, Redis (in-memory caching & presence), Schema Design, Query Optimization, B+ Trees, Sharding, ACID Transactions
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems, Multithreading, Concurrency Control, Memory Management, Computer Networks, TCP/IP, HTTP
- **Distributed & Scalable Systems:** Kafka, Event-Driven Architecture, Message Queues, Horizontal Scaling, High-Performance System Design
- **Security:** Cryptographic Hashing (bcrypt), JWT, RBAC, Audit Logging, Session Management, Secure API Design
- **DevOps & Tools:** Docker, Docker Compose, Git/GitHub, Linux, Jest, Supertest, Load Testing, Logging & Debugging

Achievements

- Felicitated with **Dean's List** award in Semester 1 for excellent academic performance.
- Achieved All India Rank **1464** in **JEE-Mains** out of over 1 million candidates and **1682** in **JEE-Advanced** 2023.
- Achieved Maharashtra State rank **67** in **MHT-CET** 2023 out of over 100000 candidates.